ISyE 6669
Lagrangian Duality Theory for Linear Optimization

Andy Sun

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Duality theory is at the heart of optimization. In this lecture, we will dive deep into the inner workings of linear programming duality. In particular, we will learn what motivates duality, why we form the dual linear program this way, and the important implications of duality. We will learn *Lagrangian duality*, which is a very general principle with applications going way beyond linear programming. This lays a foundation for our later study of duality for integer programs. We will also learn weak and strong duality for LPs and complementary slackness.

1 Example and Key Questions

Let us first start from an example. Consider the following LP:

$$
\begin{aligned}
(P) \qquad Z_P = \min \quad & x_1 + 2x_2 + 3x_3 \\
\text{s.t.} \quad & x_1 + 5x_2 + 4x_3 \geq 6 \\
& 2x_1 + 3x_2 - x_3 = 3 \\
& x_1 + x_2 - 2x_3 \leq 4 \\
& x_1 \geq 0, x_2 \leq 0, x_3 \text{ free}
\end{aligned}
$$

An introductory optimization course usually teaches the rules to form the so-called dual problem of (P) as

$$
\begin{aligned}
(D) \qquad Z_D = \max \quad & 6y_1 + 3y_2 + 4y_3 \\
\text{s.t.} \quad & y_1 + 2y_2 + y_3 \leq 1 \\
& 5y_1 + 3y_2 + y_3 \geq 2 \\
& 4y_1 - y_2 - 2y_3 = 3 \\
& y_1 \geq 0, y_2 \text{ free}, y_3 \leq 0.
\end{aligned}
$$

You can look ahead in Section 3.3 to find all the rules to form (D). But our goal is to go much beyond memorizing rules and to truly understand the rationale of duality. In particular, we will answer the following questions:

1. What motivates us to form the dual?

2. Why do we form the dual this way?

2 Motivation of taking the dual

The motivation to form the dual problem of a minimization LP (P), which is called the primal problem, is to find a *lower bound* to the optimal cost of (P). That is, we want to find a Z such that $Z \leq Z_P^*$, and in fact, we want to find the *best* lower bound possible.

This motivation may sound quite "out of the blue" — why do we want to find a lower bound to Z_P^*, if we can easily solve the primal problem to get the optimal cost? We can think about three reasons now.

1. The principle behind a systematic approach for finding the best lower bound to a minimization problem goes well beyond linear programs; in fact, it is a very general principle that can also be applied to other classes of optimization problems much harder than LPs, such as integer programming (IP) problems. There, solving a primal IP is not a trivial task at all, and the information about a good lower bound is very valuable. The principle we will learn for LP can be directly applied to IPs and also nonlinear programming problems, so studying how to find the best lower bound to an LP is a good starting point.

2. Forming an optimization problem closely related to the primal one gives us a completely new perspective to look at the original problem. Besides the lower bound, it also provides other useful information about the primal such as sensitivity, which has enormous applications in economics and engineering.

3. From an aesthetic point of view, duality is a source of symmetry and beauty. You will appreciate it as you learn more about optimization.

You may also wonder, in addition to finding a lower bound to Z_P^*, what about finding an upper bound? Notice that for a minimization problem, any feasible solution $\boldsymbol{x}$ of the primal problem gives an *upper* bound to Z_p^*. In other words, finding an upper bound is not that difficult as long as we can find any feasible solution, while finding a lower bound requires to go beyond the primal problem itself.

3 Why do we take the dual this way? – Lagrangian relaxation and Lagrangian duality

We take three steps to formulate the problem that will give the best lower bound to the optimal cost of (P). These three steps form a general procedure to formulate a dual problem for any optimization problem. The underlying general principle is called **relaxation**, more specifically, we will learn **Lagrangian relaxation**, which is a special but very important way to do relaxation.
To relax a minimization problem (P) involves three principal steps:

1. Relax the objective function: by constructing a new objective function that is always smaller or equal to the original objective function of (P) over the feasible region of (P).

2. Relax the feasible region of (P): by enlarging the feasible region of the original problem (P).

   These are the two general principles of relaxation for any optimization problem. Lagrangian relaxation specifies how to construct the new objective function and how to enlarge the feasible region. The result of carrying out Lagrangian relaxation on the primal LP is a new optimization problem, which we call the *Lagrangian relaxation problem* of the primal LP. This new LP gives a lower bound to the primal LP, and the lower bound depends on a set of parameters called the **Lagrangian multipliers**, which we use to form the Lagrangian relaxation.

3. The final step is to maximize the lower bound over these Lagrangian multipliers so that we get the best lower bound. This leads to a maximization problem, and it turns out to be an LP too. This final maximization problem is called the **Lagrangian dual** problem, or the dual problem for short.

OK. Enough words, let's see some action!

3.1 Lagrangian relaxation and Lagrangian dual

Now we discuss in detail the two principal steps of Lagrangian relaxation and the final step that forms the Lagrangian dual.

Step 1: Relax the objective function.
By relaxing the objective function, we mean to formulate a new objective function that is *smaller than or equal to* the original primal objective function, because we are minimizing. The way we do this is as follows. Take a multiple of each constraint and add it from the objective. Specifically, consider the following objective:

$$x_1 + 2x_2 + 3x_3 + y_1 \cdot (6 - x_1 - 5x_2 - 4x_3) + y_2 \cdot (3 - 2x_1 - 3x_2 + x_3) + y_3 \cdot (4 - x_1 - x_2 + 2x_3) \tag{1}$$

where y_1 and y_2 and y_3 are the *Lagrangian multipliers* chosen *so that each product term is* ≤ 0 *for any primal feasible* $\boldsymbol{x}$. In other ways, we choose y_1, y_2, y_3 so that

$$y_1 \cdot (6 - x_1 - 5x_2 - 4x_3) \leq 0 \quad \text{for any primal feasible } x_1, x_2, x_3 \tag{2a}$$
$$y_2 \cdot (3 - 2x_1 - 3x_2 + x_3) \leq 0 \quad \text{for any primal feasible } x_1, x_2, x_3 \tag{2b}$$
$$y_3 \cdot (4 - x_1 - x_2 + 2x_3) \leq 0 \quad \text{for any primal feasible } x_1, x_2, x_3. \tag{2c}$$

If the three product terms satisfy the conditions in (2a)-(2c), then the new objective function (1) is smaller than the original objective for any primal feasible solution $\boldsymbol{x}$.

Now, since any primal feasible solution $\boldsymbol{x} = (x_1, x_2, x_3)$ satisfies $x_1+5x_2+4x_3-6 \geq 0$, therefore, to ensure (2a), we only need to choose $y_1 \geq 0$. Similarly, since $2x_1 + 3x_2 - x_3 - 3 = 0$ for any primal feasible $\boldsymbol{x}$, to ensure (2b), y_2 can be either positive or negative, so y_2 is free of bounds. Finally, since $x_1 + x_2 - 2x_3 - 4 \leq 0$, to ensure (2c), we should have $y_3 \leq 0$.

To summarize, we have

$$\begin{aligned}
x_1 + 5x_2 + 4x_3 \geq 6 &\;\rightarrow\; y_1 \geq 0 \ \text{(because we want } y_1 \cdot (6 - x_1 - 5x_2 - 4x_3) \leq 0\text{)} \\
2x_1 + 3x_2 - x_3 = 3 &\;\rightarrow\; y_2 \text{ free} \ \text{(because we want } y_2 \cdot (3 - 2x_1 - 3x_2 + x_3) \leq 0\text{)} \\
x_1 + x_2 - 2x_3 \leq 4 &\;\rightarrow\; y_3 \leq 0 \ \text{(because we want } y_3 \cdot (4 - x_1 - x_2 + 2x_3) \leq 0\text{).}
\end{aligned}$$

Now we can formulate an intermediate LP with the new objective that we just formulated and all the constraints of the original primal problem (P).

$$(B) \qquad \hat{Z}(\boldsymbol{y}) = \min \quad (x_1 + 2x_2 + 3x_3) + y_1 \cdot (6 - x_1 - 5x_2 - 4x_3) + y_2 \cdot (3 - 2x_1 - 3x_2 + x_3)$$

$$
\begin{aligned}
&\quad + y_3 \cdot (4 - x_1 - x_2 + 2x_3)\\
\text{s.t.}\quad & x_1 + 5x_2 + 4x_3 \geq 6\\
& 2x_1 + 3x_2 - x_3 = 3\\
& x_1 + x_2 - 2x_3 \leq 4\\
& x_1 \geq 0, x_2 \leq 0, x_3 \text{ free.}
\end{aligned}
$$

Since for every feasible x_1, x_2, x_3, the objective of the above problem is less than or equal to the original objective of the primal problem, then the optimal cost of the above problem is less than or equal to the optimal cost of the primal problem.

It is still not clear how we can deal with (B). It seems as hard to solve as the primal problem (P). The next step is to make a new problem which is easier to solve and still gives a lower bound on Z_P.

Step 2: Relax the primal constraints.
The three linear constraints of the original primal problem complicate the problem (B). If we remove them from (B), the feasible region of (B) will be enlarged, therefore, the resulting objective function will be lowered, thus, also lower than or equal to Z_P.

Here we applied the second principle of relaxation — *enlarging the feasible region of an optimization problem by removing constraints.*

The relaxed problem becomes

$$
\begin{aligned}
(LR)\quad Z(\boldsymbol{y}) = \min_{x_1,x_2,x_3}\quad & (x_1 + 2x_2 + 3x_3) + y_1 \cdot (6 - x_1 - 5x_2 - 4x_3) + y_2 \cdot (3 - 2x_1 - 3x_2 + x_3)\\
& + y_3 \cdot (4 - x_1 - x_2 + 2x_3)\\
\text{s.t.}\quad & x_1 \geq 0, x_2 \leq 0, x_3 \text{ free.}
\end{aligned}
$$

This problem (LR) is called the **Lagrangian relaxation problem** of the original primal problem (P). The objective function of (LR) is called the **Lagriangian function**.

As we argued above, we have $\underline{Z(y) \leq \hat{Z}(y) \leq Z_P^* \text{ for any } y_1 \geq 0,\ y_2 \text{ free, and } y_3 \leq 0}$.

Furthermore, we can explicitly solve the Lagrangian relaxation problem (LR). How? Well, (LR) can be *separated* into three independent problems, each is an LP with a single variable x_j and a sign constraint on x_j. That is,

$$
\begin{aligned}
Z(\boldsymbol{y}) = & \min_{x_1:x_1\geq 0} \{(1 - (y_1 + 2y_2 + y_3))\, x_1\} + \min_{x_2:x_2\leq 0} \{(2 - (5y_1 + 3y_2 + y_3))x_2\}\\
& + \min_{x_3:x_3\text{free}} \{(3 - (4y_1 - y_2 - 2y_3))x_3\} + (6y_1 + 3y_2 + 4y_3).
\end{aligned}
$$

This is a very important structure called **separable** problems. That is, the entire optimization problem can be decomposed into solving subproblems over each individual variables. The tremendous benefit of the separable structure is that now each smaller problem can be exactly solved by hand in closed form. *In fact, the computation power of the Lagrangian relaxation approach lies exactly at here that we come from a potentially very hard primal problem to a new problem which can be decomposed into much smaller problems, each of which is solvable either by hand or by computer solvers in a much faster speed than the primal problem!*

Now let's solve the three subproblems of the Lagrangian relaxation problem.

$$\min_{x_1 : x_1 \geq 0} \{(1 - (y_1 + 2y_2 + y_3))x_1\} = \begin{cases} 0 & \text{if } 1 - (y_1 + 2y_2 + y_3) \geq 0 \quad \text{(a)} \\ -\infty & \text{otherwise} \end{cases}$$

$$\min_{x_2 : x_2 \leq 0} \{(2 - (5y_1 + 3y_2 + y_3))x_2\} = \begin{cases} 0 & \text{if } 2 - (5y_1 + 3y_2 + y_3) \leq 0 \quad \text{(b)} \\ -\infty & \text{otherwise} \end{cases}$$

$$\min_{x_3 : x_3 \text{ free}} \{(3 - (4y_1 - y_2 - 2y_3))x_3\} = \begin{cases} 0 & \text{if } 3 - (4y_1 - y_2 - 2y_3) = 0 \quad \text{(c)} \\ -\infty & \text{otherwise.} \end{cases}$$

Putting things together, the optimal cost of the Lagrangian relaxation problem is given as

$$Z(\boldsymbol{y}) = \begin{cases} 6y_1 + 3y_2 + 4y_3 & \text{if (a), (b), (c) hold} \\ -\infty & \text{otherwise.} \end{cases}$$

What this tells us is that if the conditions (a), (b), and (c) on y_1, y_2, y_3 hold, then $Z(y)$ is equal to $6y_1 + 3y_2 + 4y_3$. Furthermore, if $y_1 \geq 0$, y_2 free, $y_3 \leq 0$, then $Z(y)$ is a lower bound on Z_P, i.e. $6y_1 + 3y_2 + 4y_3 \leq Z_P$. Of course, if any of the three conditions does not hold, $Z(\boldsymbol{y}) = -\infty$, which still provides a lower bound on Z_P, but not useful at all.

The final step to form the Lagrangian dual is to find the *best* lower bound by choosing proper values for y_1, y_2, y_3.

Step 3: Finding the best lower bound.
The best lower bound means the lower bound that is the closest to Z_P. The following LP finds the best lower bound by maximizing $Z(\boldsymbol{y})$ over the constraints (a), (b), (c), and the sign constraints on y's.

$$\begin{aligned} Z_D = \max_{y_1, y_2, y_3} \quad & Z(\boldsymbol{y}) && \text{(3a)} \\ \text{s.t.} \quad & y_1 \geq 0, y_2 \text{ free}, y_3 \leq 0, && \text{(3b)} \end{aligned}$$

which can be written out as

$$\begin{aligned} (D) \quad Z_D = \max_{y_1, y_2, y_3} \quad & 6y_1 + 3y_2 + 4y_3 && \text{(4a)} \\ \text{s.t.} \quad & 1 - (y_1 + 2y_2 + y_3) \geq 0 && \text{(4b)} \\ & 2 - (5y_1 + 3y_2 + y_3) \leq 0 && \text{(4c)} \\ & 3 - (4y_1 - y_2 - 2y_3) = 0 && \text{(4d)} \\ & y_1 \geq 0, y_2 \text{ free}, y_3 \leq 0. && \text{(4e)} \end{aligned}$$

Notice that (4) is nothing but the dual LP on page 1.

3.2 Dual of the dual is the primal, at least for LP

Now let us push further and take the dual of the dual problem (4). We carry out the same steps as in the previous subsection. The differences are that (4) is maximization.

Step 1: Form the Lagrangian function and determine the sign of the Lagrangian multipliers.
Form the Lagrangian function as

$$L(\boldsymbol{x}, \boldsymbol{y}) = (6y_1 + 3y_2 + 4y_3) + x_1(1 - (y_1 + 2y_2 + y_3)) + x_2(2 - (5y_1 + 3y_2 + y_3)) + x_3(3 - (4y_1 - y_2 - 2y_3)). \tag{5}$$

We want the three product terms in the above Lagrangian function to be *nonnegative* so that the Lagrangian function will provide an *upper bound* to the original problem (D). Therefore, given the directions of inequalities (4b)-(4d), we need $x_1 \geq 0$, $x_2 \leq 0$, and x_3 free.

Step 2: Form the Lagrangian relaxation problem.
The Lagrangian relaxation problem is formed as

$$\begin{aligned} \tilde{Z}_{LR}(\boldsymbol{x}) = \max_{\boldsymbol{y}} \quad & (6y_1 + 3y_2 + 4y_3) + x_1(1 - (y_1 + 2y_2 + y_3)) + x_2(2 - (5y_1 + 3y_2 + y_3)) \\ & + x_3(3 - (4y_1 - y_2 - 2y_3)) \\ \text{s.t.} \quad & y_1 \geq 0, y_2 \text{ free}, y_3 \leq 0. \end{aligned}$$

This is a separable problem and can be solved by maximizing over each y_i separately as follows.

$$\begin{aligned} \tilde{Z}_{LR}(\boldsymbol{x}) = & \max_{y_1 \geq 0, y_3 \leq 0} (x_1 + 2x_2 + 3x_3) + y_1 \cdot (6 - (x_1 + 5x_2 + 4x_3)) + y_2 \cdot (3 - (2x_1 + 3x_2 - x_3)) \\ & \qquad + y_3 \cdot (4 - (x_1 + x_2 - 2x_3)) \qquad (6) \\ = & (x_1 + 2x_2 + 3x_3) + \max_{y_1 \geq 0} (y_1 \cdot (6 - (x_1 + 5x_2 + 4x_3))) + \max_{y_2 \text{ free}} (y_2 \cdot (3 - (2x_1 + 3x_2 - x_3))) \\ & \qquad + \max_{y_3 \leq 0} (y_3 \cdot (4 - (x_1 + x_2 - 2x_3))) \qquad (7) \\ = & \begin{cases} (x_1 + 2x_2 + 3x_3) & \text{if } 6 - (x_1 + 5x_2 + 4x_3) \leq 0,\ 3 - (2x_1 + 3x_2 - x_3) = 0,\ 4 - (x_1 + x_2 - 2x_3) \geq 0 \\ +\infty & \text{o.w.} \end{cases} \qquad (8) \end{aligned}$$

Step 3: Find the best upper bound.
We need to minimize $\tilde{Z}_{LR}(\boldsymbol{x})$ as below, which gives the primal problem (P) on page 1.

$$\begin{aligned} & \min_{x_1 \geq 0, x_2 \leq 0, x_3 \text{ free}} \tilde{Z}_{LR}(\boldsymbol{x}) \\ = & \min_{\boldsymbol{x}} \ x_1 + 2x_2 + 3x_3 \\ & \text{s.t. } 6 - (x_1 + 5x_2 + 4x_3) \leq 0 \\ & \quad 3 - (2x_1 + 3x_2 - x_3) = 0 \\ & \quad 4 - (x_1 + x_2 - 2x_3) \geq 0 \\ & \quad x_1 \geq 0, x_2 \leq 0, x_3 \text{ free}. \end{aligned}$$

3.3 General rules to form the dual problem for linear programs

From the above derivation, we can see that, to formulate the dual of a minimization LP, you only need to remember two rules: $y_i(b_i - \boldsymbol{a}_i^\top \boldsymbol{x}) \le 0$ for all i and $(c_j - \boldsymbol{y}^\top \boldsymbol{A}_j)x_j \ge 0$ for all j. Then the rules for determining the signs of the dual variables and the constraints are:

1. Since $y_i(b_i - \boldsymbol{a}_i^\top \boldsymbol{x}) \le 0$:
   - If $\boldsymbol{a}_i^\top \boldsymbol{x} \ge b_i$, then $y_i \ge 0$;
   - If $\boldsymbol{a}_i^\top \boldsymbol{x} \le b_i$, then $y_i \le 0$;
   - If $\boldsymbol{a}_i^\top \boldsymbol{x} = b_i$, then y_i free.
2. Since $(c_j - \boldsymbol{y}^\top \boldsymbol{A}_j)x_j \ge 0$:
   - If $x_j \ge 0$, then $c_j \ge \boldsymbol{y}^\top \boldsymbol{A}_j$ (i.e. $\boldsymbol{y}^\top \boldsymbol{A}_j \le c_j$);
   - If $x_j \le 0$, then $c_j \le \boldsymbol{y}^\top \boldsymbol{A}_j$ (i.e. $\boldsymbol{y}^\top \boldsymbol{A}_j \ge c_j$);
   - If x_j free, then $c_j = \boldsymbol{y}^\top \boldsymbol{A}_j$ (i.e. $\boldsymbol{y}^\top \boldsymbol{A}_j = c_j$).

To put it more generally for any form of LP. Consider first the primal LP:

$$
\begin{aligned}
\min\ & \sum_{j=1}^{n} c_j x_j && \text{(9a)}\\
\text{s.t.} & \sum_{j=1}^{n} a_{ij} x_j \ge b_i \quad \text{for all } i = 1, ..., m_1 && \text{(9b)}\\
& \sum_{j=1}^{n} a_{ij} x_j = b_i \quad \text{for all } i = m_1 + 1, ..., m_2 && \text{(9c)}\\
& \sum_{j=1}^{n} a_{ij} x_j \le b_i \quad \text{for all } i = m_2 + 1, ..., m && \text{(9d)}\\
& x_j \ge 0 \quad \text{for all } j = 1, ..., n_1 && \text{(9e)}\\
& x_j \text{ is free} \quad \text{for all } j = n_1 + 1, ..., n_2 && \text{(9f)}\\
& x_j \le 0 \quad \text{for all } j = n_2 + 1, ..., n. && \text{(9g)}
\end{aligned}
$$

Then the dual is:

$$
\begin{aligned}
\max\ & \sum_{i=1}^{m} b_i y_i && \text{(10a)}\\
\text{s.t.} & \sum_{i=1}^{m} a_{ij} y_i \le c_j \quad \text{for all } j = 1, ..., n_1 && \text{(10b)}\\
& \sum_{i=1}^{m} a_{ij} y_i = c_j \quad \text{for all } j = n_1 + 1, ..., n_2 && \text{(10c)}\\
& \sum_{i=1}^{m} a_{ij} y_i \ge c_j \quad \text{for all } i = n_2 + 1, ..., n && \text{(10d)}
\end{aligned}
$$

$$y_i \geq 0 \quad \text{for all } i = 1, ..., m_1 \tag{10e}$$
$$y_i \text{ is free} \quad \text{for all } i = m_1 + 1, ..., m_2 \tag{10f}$$
$$y_i \leq 0 \quad \text{for all } i = m_2 + 1, ..., m. \tag{10g}$$

The dual of a dual is the primal. So if the dual above (i.e. (10a) - (10g)) was the original LP, then (9a) - (9g) would be its dual.

Now, we can summarize from the above examples the general rules to form the dual problem.

Primal	Minimize	Maximize	Dual
Constraints	$\geq b_i$ $\leq b_i$ $= b_i$	≥ 0 ≤ 0 free	Variables
Variables	≥ 0 ≤ 0 free	$\leq c_j$ $\geq c_j$ $= c_j$	Constraints

Again, these rules are not for mere memorization. They should not replace our understanding of how LP duality works. If you really understand the previous three examples, you should be able to quickly derive the dual of any LP. In the beginning, you may need to go through in your mind the first-principle derivations given in the examples, but each time is a reinforcement of your understanding, until taking the dual becomes so natural and transparent.

4 Weak Duality and Strong Duality

4.1 Weak duality and its implications

Given a primal minimization problem, by the way we construct the dual, it is clear that the any dual feasible solution will give a lower bound to the primal objective cost. More precisely, we have the following result called Weak Duality.

Theorem 1 (Linear programming weak duality)**.** *If $\boldsymbol{x}$ is any feasible solution to the primal minimization LP* (9)*,and $\boldsymbol{y}$ is any feasible solution to the dual maximization LP* (10)*, then $\boldsymbol{c}^\top \boldsymbol{x} \geq \boldsymbol{b}^\top \boldsymbol{y}$.*

Weak duality is a very useful property. With the weak duality theorem at hand, we can immediately get the following implications:

- If the optimal cost of the primal minimization problem is $-\infty$, then the dual maximization problem must be infeasible.
- If the optimal cost of the dual maximization problem is $+\infty$, then the primal minimization problem must be infeasible.

Why are they true? If the optimal cost of the primal problem is $-\infty$, suppose the dual maximization problem has a feasible solution $\boldsymbol{y}$ which is a finite number, then by weak duality, we have $-\infty > \boldsymbol{b}^\top \boldsymbol{y}$, which is impossible for any finite number $\boldsymbol{y}$. Similar argument applies to the second statement.

Here is another true statement implied by the weak duality theorem.

- Let $\boldsymbol{x}^*$ be feasible to the primal problem and $\boldsymbol{y}^*$ be feasible to the dual problem, and suppose $\boldsymbol{c}^\top \boldsymbol{x}^* = \boldsymbol{b}^\top \boldsymbol{y}^*$, then $\boldsymbol{x}^*$ and $\boldsymbol{y}^*$ are optimal solutions to the primal and dual problems, respectively.

Suppose the above statement is not true. We want to derive a contradiction. So suppose $\boldsymbol{x}^*$ is not optimal for the primal problem, then there exists a primal optimal solution $\bar{\boldsymbol{x}}$ such that $\boldsymbol{c}^\top \bar{\boldsymbol{x}} < \boldsymbol{c}^\top \boldsymbol{x}^*$. But this would imply $\boldsymbol{c}^\top \bar{\boldsymbol{x}} < \boldsymbol{b}^\top \boldsymbol{y}^*$, which contradicts the weak duality theorem. Similarly, we can prove that $\boldsymbol{y}^*$ is optimal for the dual.

4.2 Strong duality and its implications

Theorem 2 (Linear programming Strong Duality)**.** *If a primal linear program has a finite optimal solution $\boldsymbol{x}^*$, then its dual linear program must also have a finite optimal solution $\boldsymbol{y}^*$, and the respective optimal objective values are equal, that is $\boldsymbol{c}^\top \boldsymbol{x}^* = \boldsymbol{b}^\top \boldsymbol{y}^*$.*

Proof. Assume the primal problem is a standard form LP (without loss of generality, we can tranform any LP to standard form). Then the primal and dual problems are:

$$
\begin{aligned}
(P) \quad \min \quad & \boldsymbol{c}^\top \boldsymbol{x} \\
\text{s.t.} \quad & \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \\
& \boldsymbol{x} \geq \boldsymbol{0}.
\end{aligned}
\qquad\qquad
\begin{aligned}
(D) \quad \max \quad & \boldsymbol{b}^\top \boldsymbol{y} \\
\text{s.t.} \quad & \boldsymbol{y}^\top \boldsymbol{A} \leq \boldsymbol{c}^\top \\
& \boldsymbol{y} \text{ free.}
\end{aligned}
$$

If the primal problem has a finite optimal solution, then the simplex method with anti-cycling rule like the Bland's rule will terminate with an optimal solution $\boldsymbol{x}^*$ and the associated basis matrix $\boldsymbol{B}$. By the optimality of $\boldsymbol{x}^*$, we know the reduced costs are $\bar{\boldsymbol{c}}^\top = \boldsymbol{c}^\top - \boldsymbol{c}_B^\top \boldsymbol{B}^{-1}\boldsymbol{A} \geq \boldsymbol{0}^\top$. Now let $(\boldsymbol{y}^*)^\top = \boldsymbol{c}_B^\top \boldsymbol{B}^{-1}$, then the reduced costs condition is the same as $(\boldsymbol{y}^*)^\top \boldsymbol{A} \leq \boldsymbol{c}^\top$, i.e. $\boldsymbol{y}^*$ is dual feasible. Also, $\boldsymbol{b}^\top \boldsymbol{y}^* = \boldsymbol{c}_B^\top \boldsymbol{B}^{-1}\boldsymbol{b} = \boldsymbol{c}_B^\top \boldsymbol{x}_B = \boldsymbol{c}^\top \boldsymbol{x}^*$. By Weak Duality, $\boldsymbol{y}^*$ is an optimal solution to the dual problem, and the respective optimal objective values of primal and dual are equal. □

This proof shows a very important link between the primal (P) and the dual (D), namely the optimal dual solution $\boldsymbol{y}^*$ can be obtained from the output of the Simplex method, i.e. $\boldsymbol{y}^* = (\boldsymbol{c}_B^\top \boldsymbol{B})^\top$.

5 The Table of Possibles and Impossibles

Use the weak and strong duality discussed in the last lecture, we can now investigate all possible outcomes of the primal and dual problems. The primal and dual problems can be *finite optimal*, or *unbounded optimal*, or *infeasible*. So, there are in total 9 combinations. Are all these 9 combinations possible? The answer is no. The duality theory puts restrictions on the possible combinations.

1. Primal *finite optimal*, Dual *finite optimal*: **Possible**. In this case, the strong duality tells us that $\boldsymbol{c}^\top \boldsymbol{x}^* = \boldsymbol{b}^\top \boldsymbol{y}^*$ for primal optimal solution $\boldsymbol{x}^*$ and dual optimum $\boldsymbol{y}^*$. The strong duality actually tells us: if Primal *finite optimal*, Dual **must** be *finite optimal*, and vice versa.

2. Primal *finite optimal*, Dual *unbounded*: **Impossible** due to the weak duality. If the dual has unbounded optimum, namely $+\infty$, it can not be a lower bound for the finite optimal cost of the primal.

3. Primal *finite optimal*, Dual *infeasible*: **Impossible** due to the strong duality, which says if the primal has a finite optimal cost, so does the dual.

4. Primal *unbounded*, Dual *finite optimal*: **Impossible** due to the weak duality. It can also be seen by symmetry with the case Primal *finite optimal*, Dual *unbounded*.

5. Primal *unbounded*, Dual *unbounded*: **Impossible** due to the weak duality. Both unbounded means that Primal optimal cost is $-\infty$, and dual optimal cost is $+\infty$, which can never happen.

6. Primal *unbounded*, Dual *infeasible*: **Possible** due to the weak duality. Primal has $-\infty$ cost, the dual cannot have any feasible solution, otherwise that feasible solution would give a lower bound to $-\infty$ by weak duality, which is clearly impossible. The weak duality actually tells us: if Primal *unbounded*, Dual **must** be *infeasible*.

7. Primal *infeasible*, Dual *finite optimal*: **Impossible** by using the result of Primal *finite optimal*, Dual *infeasible*.

8. Primal *infeasible*, Dual *unbounded*: **Possible** by using the result of Primal *unbounded*, Dual *infeasible*. But notice that when Primal *infeasible*, the dual may not necessarily be *unbounded*. In other words, it is not true that if Primal *infeasible*, the dual must be *unbounded*. The reason is exactly the next situation.

9. Primal *infeasible*, Dual *infeasible*: **Possible**. Not directly covered by weak duality or strong duality. But look at the following example.

$$
\begin{aligned}
\min \quad & x_1 + 2x_2 \\
\text{s.t.} \quad & x_1 + x_2 = 1 \\
& 2x_1 + 2x_2 = 3 \\
& x_1, x_2 \text{ free.}
\end{aligned}
\qquad\qquad
\begin{aligned}
\max \quad & y_1 + 3y_2 \\
\text{s.t.} \quad & y_1 + 2y_2 = 1 \\
& y_1 + 2y_2 = 2 \\
& y_1, y_2 \text{ free.}
\end{aligned}
$$

To summarize, we have the following **Table of Possibles and Impossibles**.

	Finite Optimum	**Unbounded**	**Infeasible**
Finite Optimum	Possible	Impossible	Impossible
Unbounded	Impossible	Impossible	Possible
Infeasible	Impossible	Possible	Possible

Notice this table is exactly symmetric, because the dual of the dual is the primal.

6 Complementary Slackness

Besides the weak and strong duality, here is another important relation between primal and dual solutions: the *complementary slackness* conditions. It is equivalent to the strong duality.

Theorem 3 (Complementary Slackness)**.** *Let* $\boldsymbol{x}$ *and* $\boldsymbol{y}$ *be feasible solutions to the primal and dual problem, respectively. Then* $\boldsymbol{x}$ *and* $\boldsymbol{y}$ *are optimal solutions for the two respective problems if and only if they satisfy the following conditions:*

Primal Complementary Slackness: $y_i(\boldsymbol{a}_i^\top \boldsymbol{x} - b_i) = 0$ *for all* i*. In words, either the* i*-th primal constraint is active (binding, tight) so* $\boldsymbol{a}_i^\top \boldsymbol{x} = b_i$*, or the corresponding dual variable* $y_i = 0$*.*

Dual Complementary Slackness: $x_j(c_j - \boldsymbol{y}^\top \boldsymbol{A}_j) = 0$ *for all* j*. In words, either the* j*-th dual constraint is active (binding, tight) so* $\boldsymbol{y}^\top \boldsymbol{A}_j = c_j$*, or the corresponding primal variable* $x_j = 0$*.*

Proof sketch. From the construction of the dual problem (See Section 3, especially the summary in the end), we know that for any primal feasible $\boldsymbol{x}$ and dual feasible $\boldsymbol{y}$, we have

$$\begin{aligned} y_i \cdot (\boldsymbol{a}_i^\top \boldsymbol{x} - b_i) &\geq 0 \quad \forall i = 1, \ldots, m \\ x_j \cdot (c_j - \boldsymbol{y}^\top \boldsymbol{A}_j) &\geq 0 \quad \forall j = 1, \ldots, n. \end{aligned}$$

Let's denote $u_i \triangleq y_i \cdot (\boldsymbol{a}_i^\top \boldsymbol{x} - b_i)$ and $v_j \triangleq x_j \cdot (c_j - \boldsymbol{y}^\top \boldsymbol{A}_j)$. Then $u_i \geq 0$ for all i, and $v_j \geq 0$ for all j. Now let us sum all u_i and v_j together:

$$\begin{aligned} \sum_{i=1}^m u_i + \sum_{j=1}^n v_j &= \sum_{i=1}^m y_i \cdot (\boldsymbol{a}_i^\top \boldsymbol{x} - b_i) + \sum_{j=1}^n x_j \cdot (c_j - \boldsymbol{y}^\top \boldsymbol{A}_j) \\ &= \sum_{j=1}^n c_j x_j - \sum_{i=1}^m b_i y_i + \sum_{i=1}^m y_i \boldsymbol{a}_i^\top \boldsymbol{x} - \sum_{j=1}^n x_j \boldsymbol{y}^\top \boldsymbol{A}_j \\ &= \boldsymbol{c}^\top \boldsymbol{x} - \boldsymbol{b}^\top \boldsymbol{y} + \boldsymbol{y}^\top \boldsymbol{A} \boldsymbol{x} - \boldsymbol{x}^\top \boldsymbol{A}^\top \boldsymbol{y} \qquad (\text{Note } \boldsymbol{y}^\top \boldsymbol{A} \boldsymbol{x} = \boldsymbol{x}^\top \boldsymbol{A}^\top \boldsymbol{y}) \\ &= \boldsymbol{c}^\top \boldsymbol{x} - \boldsymbol{b}^\top \boldsymbol{y}. \end{aligned}$$

If $\boldsymbol{x}$ is primal optimal and $\boldsymbol{y}$ is dual optimal, then by the Strong Duality theorem, $\boldsymbol{c}^\top \boldsymbol{x} - \boldsymbol{b}^\top \boldsymbol{y} = 0$. Therefore, $\sum_{i=1}^m u_i + \sum_{j=1}^n v_j = 0$. But since each term of u_i and v_j are nonnegative, for the sum to be zero, each term must be zero, i.e. $u_i = 0$ for all i, and $v_j = 0$ for all j. This proves one direction of the theorem.

For the other direction, if $u_i = 0$ for all i, and $v_j = 0$ for all j, then we have $\boldsymbol{c}^\top \boldsymbol{x} - \boldsymbol{b}^\top \boldsymbol{y} = 0$. By the Weak Duality theorem, we know that $\boldsymbol{x}$ is primal optimal, and $\boldsymbol{y}$ is dual optimal. This completes the proof. □

Here is an example of how to use the complementary slackness.

Example 1

Consider the following primal dual pair.

$$\begin{array}{llcll} \min & 13x_1 + 10x_2 + 6x_3 & \qquad\qquad & \max & 8y_1 + 3y_2 \\ \text{s.t.} & 5x_1 + x_2 + 3x_3 = 8 & & \text{s.t.} & 5y_1 + 3y_2 \leq 13 \\ & 3x_1 + x_2 = 3 & & & y_1 + y_2 \leq 10 \\ & x_1, x_2, x_3 \geq 0 & & & 3y_1 \leq 6. \end{array}$$

The primal involves three variables, while the dual has only two variables. Someone solved the dual problem graphically, and told us the dual optimal solution $y_1^* = 2, y_2^* = 1$. We want to use this information and Complementary Slackness to get the optimal solution of the primal.

1. Use Dual Complementary Slackness: At the dual optimum, we have

$$
\begin{aligned}
&5y_1^* + 3y_2^* = 13 \\
&y_1^* + y_2^* = 3 < 10 \\
&3y_1^* = 6.
\end{aligned}
$$

   Therefore, by the Dual Complementary Slackness $x_2^*(y_1^* + y_2^* - 10) = 0$, we know $x_2^* = 0$.

2. Primal Complementary Slackness is automatically satisfied, because the primal constraints are equalities. From above, we know $x_2^* = 0$, therefore we have

$$
\begin{aligned}
&5x_1^* + 3x_3^* = 8 \\
&3x_1^* = 3.
\end{aligned}
$$

   The optimal primal solution is $x_1^* = 1, x_2^* = 0, x_3^* = 1$.

Example 2

Use the same primal and dual problem of Example 1. If we are instead given a primal feasible solution $(x_1^* = 1, x_2^* = 0, x_3^* = 1)$, can we verify this is an optimal solution? Of course, we can check the reduced cost as we did in the simplex method. But we can also do it by using Complementary Slackness. In particular, if we can find a dual feasible solution that satisfies Complementary Slackness conditions with the primal solution $\boldsymbol{x}^*$, then both are optimal for their respective problem.

1. Use Dual Complementary Slackness: Since both x_1^* and x_3^* are nonzero, in order to have $x_1^*(5y_1^* + 3y_2^* - 13) = 0$, and $x_3^*(3y_1^* - 6) = 0$, we must have

$$
\begin{aligned}
&5y_1^* + 3y_2^* = 13 \\
&3y_1^* = 6,
\end{aligned}
$$

   which gives $y_1^* = 2, y_2^* = 1$. This is a feasible dual solution.

2. Check Primal Complementary Slackness: Since the primal constraints are equalities, Primal Complementary Slackness automatically holds for this pair of primal and dual solutions. Thus, by the Complementary Slackness Theorem, we know $(x_1^* = 1, x_2^* = 0, x_3^* = 1)$ and $(y_1^* = 2, y_2^* = 1)$ are primal and dual optimal.

7 Interpretations of Dual Variables as Shadow Prices

The dual variables and the dual program have a very interesting economic interpretation. This can be illustrated by the famous Diet Problem. Suppose there are n types of foods. Each food j costs c_j dollars per unit to purchase, and has a_{ij} units of nutrient i for $i = 1, ..., m$. The goal is to combine n types of food with the minimum cost to produce an ideal food that contains b_i units of nutrient i for $i = 1, ..., m$. How should we formulate this problem as a linear program? It is quite easy to come to the following model.

$$
\min \quad \sum_{j=1}^{n} c_j x_j \tag{11}
$$

$$\text{s.t.} \quad \sum_{j=1}^{n} a_{ij} x_j \geq b_i, \qquad \forall i = 1, \ldots, m, \tag{12}$$

$$x_j \geq 0, \qquad \forall j = 1, \ldots, n. \tag{13}$$

Suppose the optimal food combination is x_j^* for $j = 1, \ldots, n$. We want to study how the change of right-hand side of the constraint, namely b_i's, would affect the optimal cost of the diet problem. In particular, *if b_i of a specific nutrient i increases by a unit, while holding all other components of the RHS constant, how would the minimum cost change? Should it increase or decrease?* The answer is quite simple: the minimum cost should increase, or at least, not decrease, because the increase of b_i would make the feasible region of the diet problem *smaller*, thus the minimum cost should go up. This argument leads to the concept of *shadow price.*

Definition 1. *The shadow price of a constraint is the change of the objective cost from the current optimal level if the right-hand side of this constraint is increased by a unit from the current level.*

In the diet problem, we introduce a shadow price y_i for each constraint i, thus nutrient i. It is the increase of the optimal objective cost from the current level by increasing b_i by a unit, and $y_i \geq 0$ because the $\geq$ direction of (12). The shadow price y_i can be understood as the unit price of the nutrient i that you would be willing to purchase, or put it in another way, y_i is the value of nutrient i at the current level of the nutrient requirement b_i's.

With this interpretation, we can interpret $\sum_{i=1}^{m} b_i y_i$ as the total value of the nutrients of the ideal food, and $\sum_{i=1}^{m} a_{ij} y_i$ for food j is the total value of nutrients that a unit of food j provides.

8 Robust Optimization

Given a linear constraint

$$\boldsymbol{a}^\top \boldsymbol{x} \leq b, \tag{14}$$

suppose the coefficient $\boldsymbol{a}$ is not known exactly, but all we know is that $\boldsymbol{a}$ lives in some polytope $\mathcal{U}$

$$\mathcal{U} = \{\boldsymbol{a} \in \mathbb{R}^n : \boldsymbol{B}\boldsymbol{a} \leq \boldsymbol{d}\}. \tag{15}$$

Then, it does not make much sense if we only obtain a solution $\hat{\boldsymbol{x}}$ to (14) for a specific value of $\hat{\boldsymbol{a}}$ in $\mathcal{U}$, because such a solution $\hat{\boldsymbol{x}}$ may not be feasible if coefficient $\boldsymbol{a}$ takes other value in $\mathcal{U}$, i.e. even if $\hat{\boldsymbol{a}}^\top \hat{\boldsymbol{x}} \leq b$, it does not imply $\boldsymbol{a}^\top \hat{\boldsymbol{x}} \leq b$ for $\boldsymbol{a} \neq \hat{\boldsymbol{a}}$. So, we need to have a different framework to make decisions in this situation when the coefficients of the constraints are uncertain. This situation is called *decision making under uncertainty.* It is a very interesting, useful, and active area of research and practice. We will study a very new paradigm for decision making under uncertainty, called robust optimization.

The idea of robust optimization is very simple. In the above situation of uncertain coefficient $\boldsymbol{a}$, instead of just considering one specific value of $\boldsymbol{a}$, we will make a decision $\boldsymbol{x}$ such that it remains feasible for all variations of $\boldsymbol{a}$ in $\mathcal{U}$. So, we have the following problem: Find $\boldsymbol{x}$ such that

$$\boldsymbol{a}^\top \boldsymbol{x} \leq b, \quad \forall \boldsymbol{a} \in \mathcal{U}. \tag{16}$$

We call such a constraint *robust constraint*. Notice that since $\mathcal{U}$ could contain infinite number of realizations of $\boldsymbol{a}$, the robust constraint (16) actually contains the conjunction of an infinite number of linear constraints, each for a specific value of $\boldsymbol{a}$ in $\mathcal{U}$. In order to deal with (16), we need to reformulate it into a finite set of constraints that we can handle.

The following procedure gives a general way to reformulate the robust constraint (16) into a set of usual linear constraints

$$
\begin{aligned}
& \boldsymbol{a}^\top \boldsymbol{x} \le b, \quad \forall \boldsymbol{a} \in \mathcal{U} := \{\boldsymbol{a} \in \mathbb{R}^n : \boldsymbol{B}\boldsymbol{a} \le \boldsymbol{d}\}, \\
\Leftrightarrow \quad & \max_{\boldsymbol{a}:\boldsymbol{B}\boldsymbol{a}\le\boldsymbol{d}} \boldsymbol{a}^\top \boldsymbol{x} \le b, && \text{(17a)} \\
\Leftrightarrow \quad & \min_{\boldsymbol{y}:\boldsymbol{B}^\top\boldsymbol{y}=\boldsymbol{x},\boldsymbol{y}\ge\boldsymbol{0}} \boldsymbol{d}^\top \boldsymbol{y} \le b, && \text{(17b)} \\
\Leftrightarrow \quad & \begin{cases} \boldsymbol{d}^\top \boldsymbol{y} \le b, \\ \boldsymbol{B}^\top \boldsymbol{y} = \boldsymbol{x}, \\ \boldsymbol{y} \ge \boldsymbol{0}. \end{cases} && \text{(17c)}
\end{aligned}
$$

Eq. (17a) uses a simple observation that if a real number (i.e. b) is greater than or equal to a set of real numbers (i.e. $\boldsymbol{a}^\top \boldsymbol{x}$ for each $\boldsymbol{a} \in \mathcal{U}$), then b should be greater than or equal to the supremum of this set of real numbers; moreover, the supremum is achieved by some particular $\boldsymbol{a}$ in $\mathcal{U}$ because the function $\boldsymbol{a}^\top \boldsymbol{x}$ and $\mathcal{U}$ are linear, so the max operator and the supremum operator coincide. Eq. (17b) holds because the uncertainty set $\mathcal{U}$ is a nonempty polytope, so the maximum in (17a) for any $\boldsymbol{x}$ is finite, therefore, LP strong duality holds between the primal maximization problem and the dual problem in (17b). Finally, since the minimum in (17b) exists for any $\boldsymbol{x}$ and is less than or equal to b, then there must exist some $\boldsymbol{y}$ in the dual polyhedron so that the objective value of the dual problem is less than b.

This above procedure of reformulating the robust linear constraint is a general procedure that can be applied to any polyhedral uncertainty set.